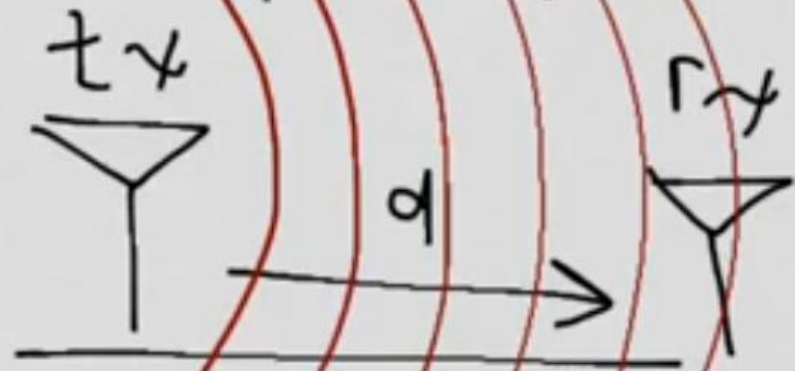


Free-space path loss (Section 1.2)

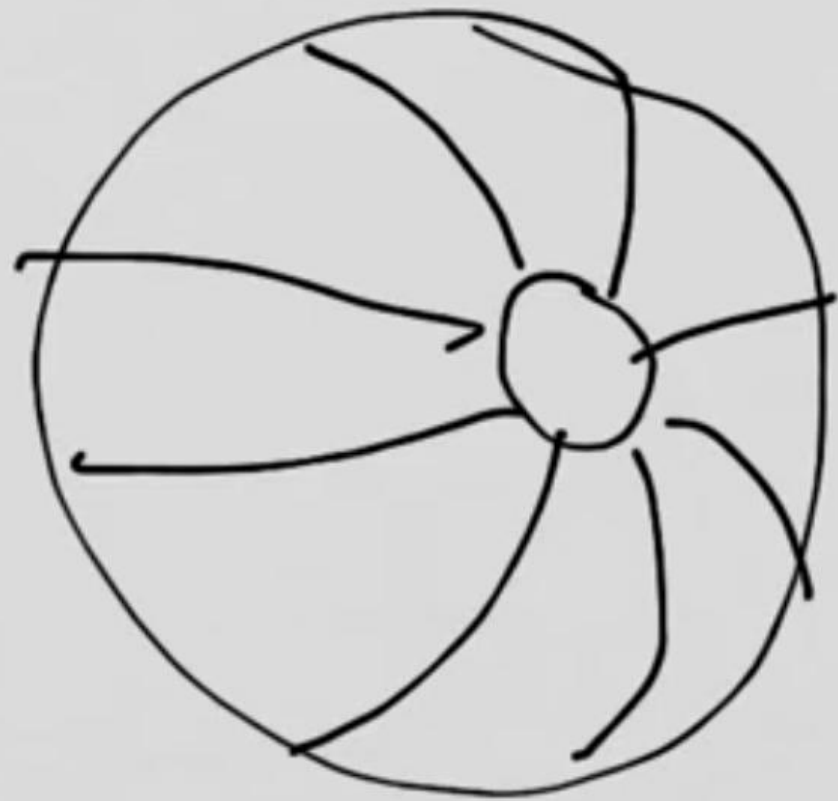


$d \sim \text{distance} \sim \text{m}$

$P_t \sim \text{power of } t_x \sim \text{W} \sim \text{mW}$

$P_r \sim \text{power of } r_x \sim \text{W} \sim \text{mW}$

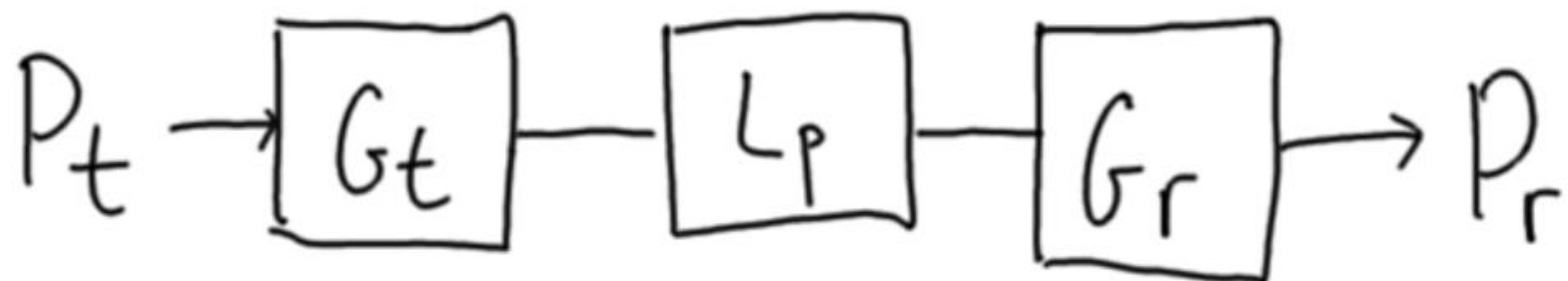
$\lambda \sim \text{wavelength} \sim \text{m}$



$f \sim \text{frequency} \sim \text{Hz}$

$$\lambda f = c = 3 \times 10^8 \text{ m/s}$$

$$\left. \begin{array}{l} f \sim \text{frequency} \sim \text{Hz} \\ \lambda f = c = 3 \times 10^8 \text{ m/s} \end{array} \right\} \lambda = \frac{c}{f}$$



$$10W \quad \times 5 \quad \div 25 \quad \times 3 \quad P_r = 6W$$

$$P_r = \frac{10W \times 5 \times 3}{25}$$

$$P_r = \frac{P_t \cdot G_t \cdot G_r}{L_p}$$

$$P_r = A_e \Phi$$

→ effective antenna area m^2

$\Phi \sim$ power flux density



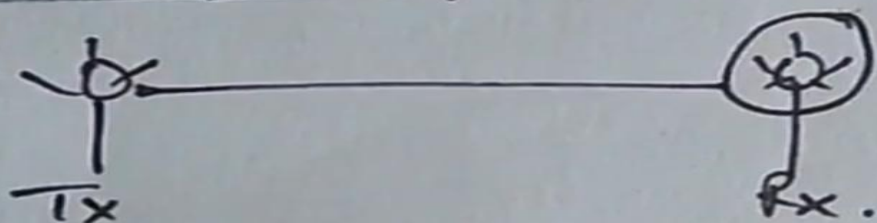
$$\Phi = \frac{\text{Power}}{\text{Surface area}} \sim \frac{W}{m^2}$$

$$P_r = A_e \frac{P_t}{4\pi d^2}$$

$$G = \frac{4\pi A_e}{\lambda^2}$$

$$\hookrightarrow A_e = \left(\frac{\lambda^2 G}{4\pi} \right)$$

Receiver Section



$A_e \uparrow$ $\rightarrow$ More is the radiation absorbed by the Receiver.

$$P_D = \frac{P_T}{4\pi R^2} \cdot G \cdot K.$$

$$G = \frac{4\pi A}{\lambda^2} \quad (\text{Antenna Theory}).$$

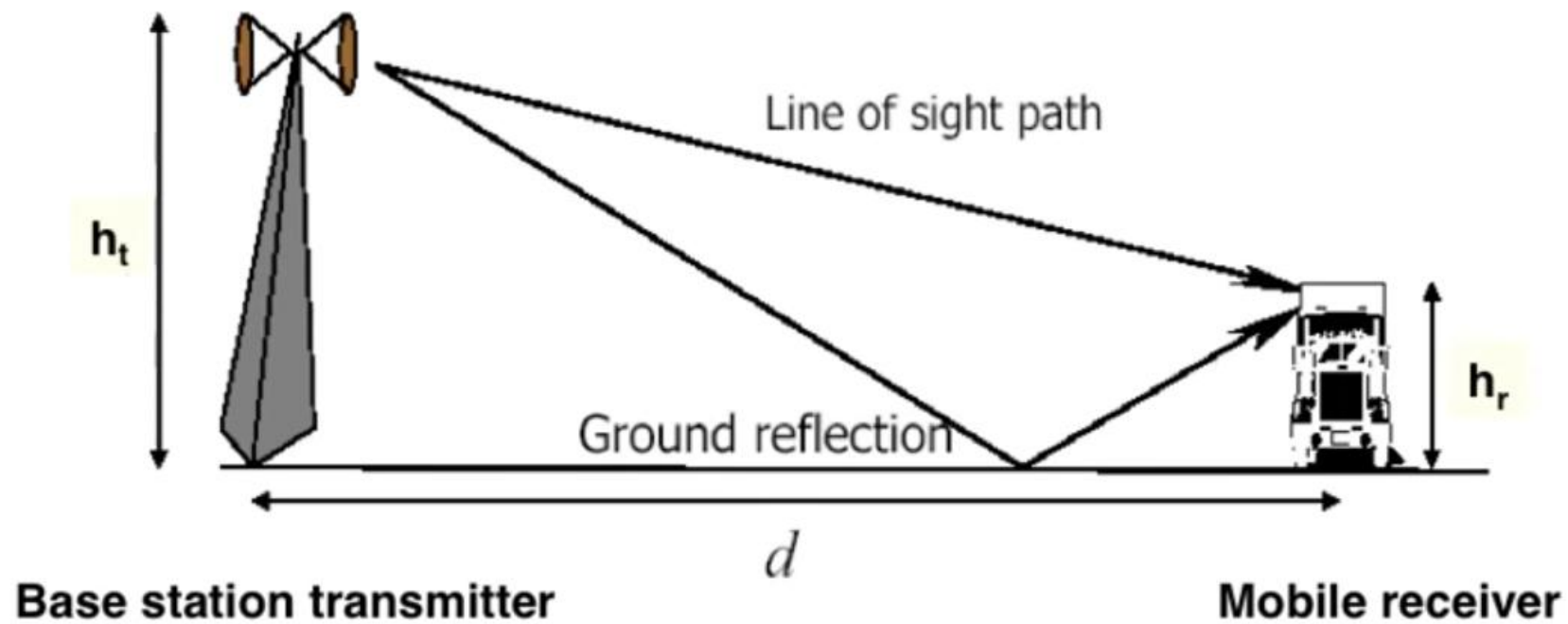
$A \rightarrow$ Effective Aperture of Receiver

$\lambda =$ Wavelength.

$$A = \frac{G \lambda^2}{4\pi}.$$

$$P_R = P_T G_T \cdot G_R \cdot K \left[\frac{\lambda^2}{4\pi d^2} \right].$$

Two-Ray Propagation Model



Two Ray Propagation Model

$$P_r = \frac{P_t G_t G_r h_t^2 h_r^2}{d^4}$$

Holds for $d \gg \sqrt{h_t h_r}$

P_r = received power
 P_t = transmitted power
 G_t = transmitter antenna gain
 G_r = receiver antenna gain
 h_t = transmitter height
 h_r = receiver height
 d = transmitter-receiver separation distance

$$\text{Path Loss in dB} = 10 \log P_r/P_t$$

Common simplification

$$P_r(d_r) = P_r(d_0) (d_0/d_r)^n$$

n : 2 to 4 in urban cellular systems



$$P_r(d) = P_r(d_0) (d_0/d)^4$$

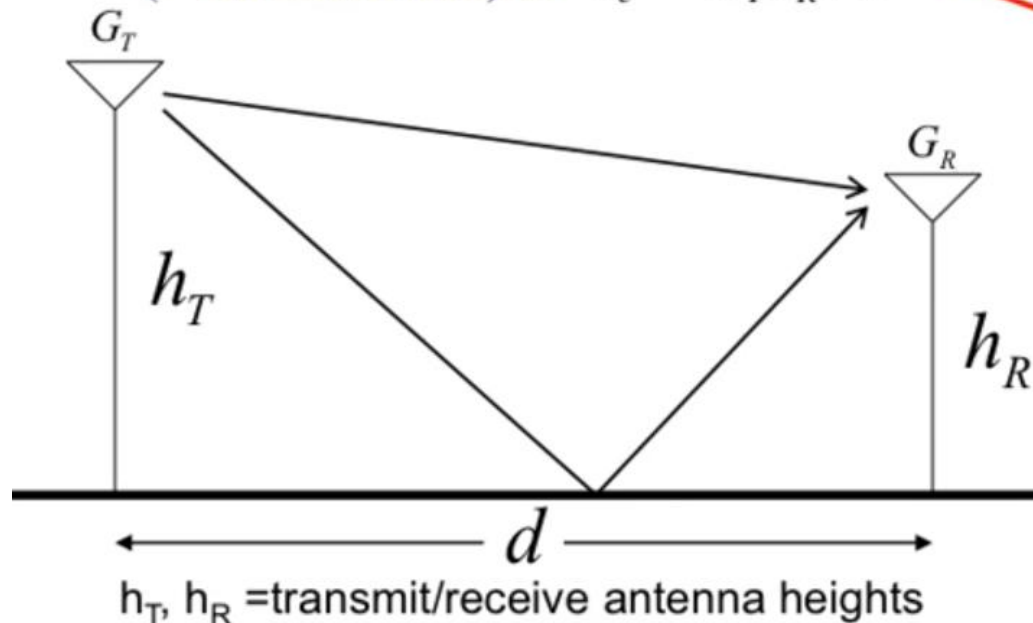
Two ray -model

- Also called a "Reflection + Plane Earth" –model
- One LOS path and one reflected path (i.e., a single ground reflection dominates the multipath effect)
- The received power is given by $P_R = P_T G / L$

where the path loss is $L = \frac{1}{\left(4 \sin^2 \left(\frac{2\pi h_T h_R}{\lambda d} \right) \left(\frac{\lambda}{4\pi d} \right)^2 \right)}$

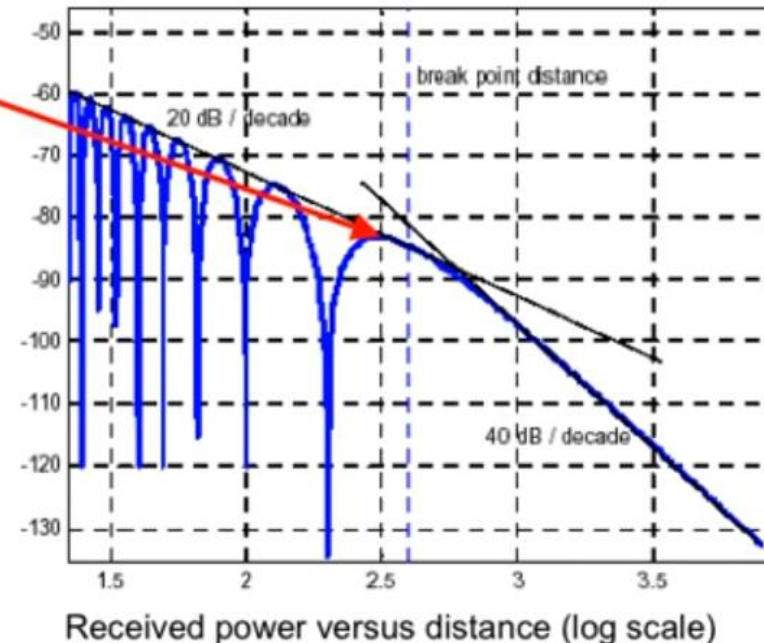
- Break-point (cross over) distance

($\approx$ last maximum) is: $d_c \approx 4h_T h_R / \lambda$



At low distances these two paths are either constructive or destructive depending on the relative path length difference $\rightarrow$ fading signal

When the path difference decreases (distance increases) the sum signal approaches to zero, because of phase reversal at the reflection.



Two ray –model (cont'd)

□ Assuming $\lambda d \gg h_T h_R$, the received power can be approximated based on the fact that $\sin x \approx x$ when x is close to zero.

$$\frac{1}{L} = 4 \sin^2 \left(\frac{2\pi h_T h_R}{\lambda d} \right) \left(\frac{\lambda}{4\pi d} \right)^2 \approx 4 \left(\frac{2\pi h_T h_R}{\lambda d} \right)^2 \left(\frac{\lambda}{4\pi d} \right)^2 = \left(\frac{h_T h_R}{d^2} \right)^2 \Rightarrow L = \frac{d^4}{(h_T h_R)^2}$$

□ This is so called "Plane Earth" propagation equation, which differs from the free-space equation in three ways:

- Power falls off **independently of frequency**
- It shows an **inverse fourth-power law** (at the limit, when d is sufficiently high), rather than inverse square-law of free space propagation. It means a far more rapid attenuation of the received power.
- It shows the **effect of the transmit and receive antenna heights** on propagation losses.

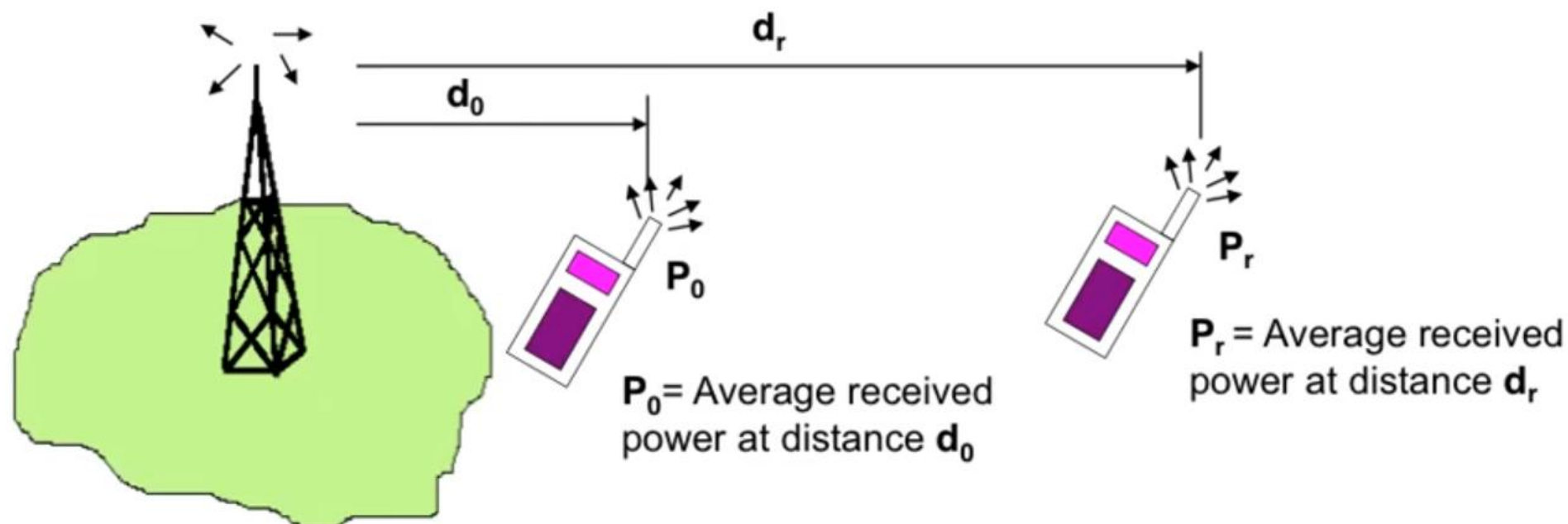
□ For $d < d_c$ use free space model, for $d > d_c$ use 2-ray model

□ *The two-ray model is realistic for so called LOS (line-of-sight) microcells.*

General ray-tracing models

- The 2-path model requires information on antenna heights only, but it is not suitable for environments with many reflectors, such as urban areas and indoor environments
- More general analytical models with $N > 2$ rays have been also derived.
- In practice, ray-tracing models utilize extensive numerical computations, using efficient algorithms, to evaluate the significant rays from the knowledge of antenna positions, streets/hallways geometry, the geometry and dielectric properties of reflectors, diffractors and scatterers in the environment.
 - Detailed knowledge of the environment and heavy computations are needed.

Simplified Mobile Channel Propagation Model



Received signal power P_r is proportional to d_r^{-n} , where n is the **path loss exponent**
In urban cellular systems n ranges from 2 to 4

Simplified path loss model

- In the simplified path loss model, the received power is:

$$P_R(d) = P_R(d_0) \left(\frac{d_0}{d} \right)^n$$

Determined
empirically

d_0 - a reference distance (e.g., 1 m)

n - path loss exponent

- Not very accurate. Wide range of empirical path-loss exponents, especially indoors, due to attenuations caused by the geometry of the environment.

- Without knowledge of the actual environment, the path loss cannot be predicted more accurately.

Environment	n
free space	2
flat rural, home	3
rolling rural	3.5
suburban	4
dense urban	4.5
Office building (same floor)	1.6-3.5
Office building (multiple floors)	2-6

4 Decibels and Signal Strength

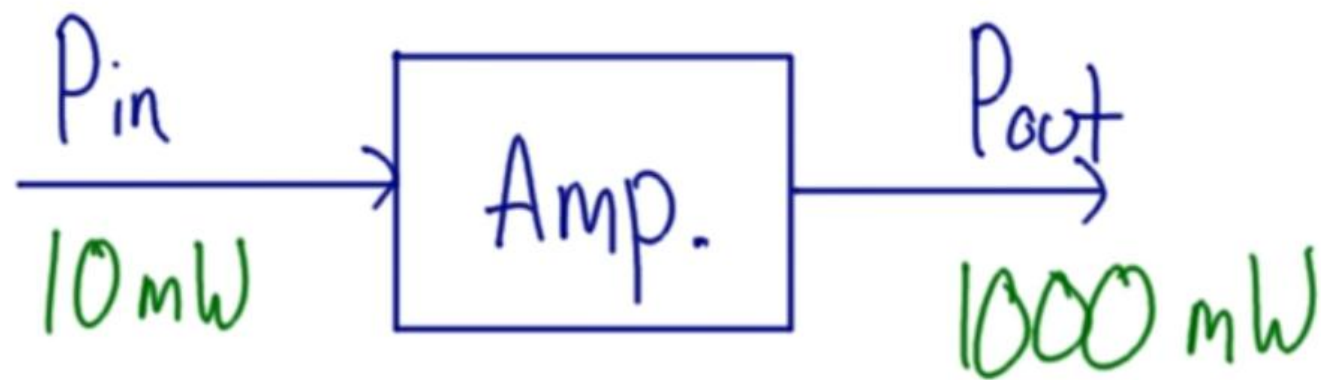
In a transmission system a signal is attenuated as it propagates along a transmission medium. That is, there is a *loss* of signal strength. To compensate, amplifying components may be used, producing a *gain* in signal strength. *Decibels* are commonly used to express gains and losses in a transmission system.

A decibel is a measure of the ratio between two signal levels:

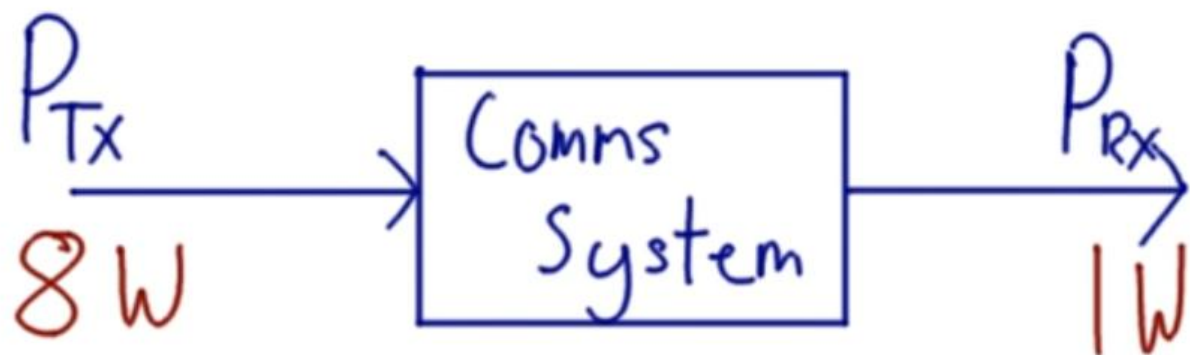
$$G_{\text{dB}} = 10 \log_{10} \frac{P_{\text{out}}}{P_{\text{in}}}$$

where:

- G_{dB} is the gain, in decibels
- P_{in} is the input power level



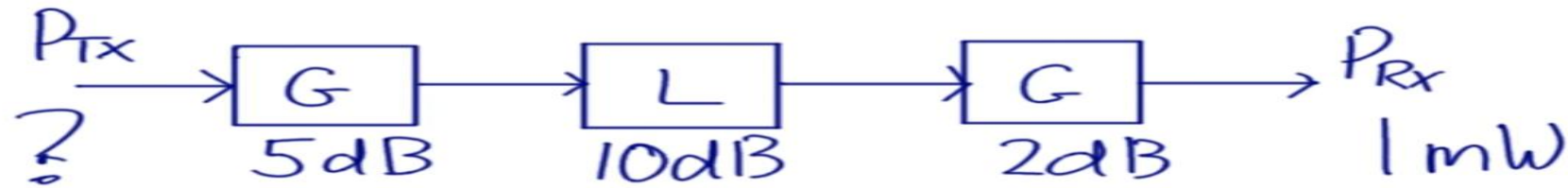
$$\text{Gain} = \frac{P_{out}}{P_{in}} = \frac{1000\text{ mW}}{10\text{ mW}} = 100$$
$$= 20\text{ dB}$$





$$\text{Loss} = \frac{P_{Tx}}{P_{Rx}} = \frac{8W}{1W} = 8 = 9\text{dB}$$

$$\text{Gain} = \frac{P_{Rx}}{P_{Tx}} = \frac{1W}{8W} = 0.125 = -9\text{dB}$$



$$\begin{aligned} \text{System gain} &= 5\text{dB} - 10\text{dB} + 2\text{dB} \\ &= -3\text{dB} \end{aligned}$$

$$\text{System loss} = 3\text{dB}$$

$$P_{Tx} = 2\text{mW}$$